# Feature Extraction Techniques for the Classification of Four-Class Motor Imagery Based EEG Data: A Comparison

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**Abstract.** In this paper, the four-class classification (Left hand, right hand, feet, and tongue) Motor Imagery (MI) signals is performed using four different feature extraction techniques. First, raw EEG signals are pre-processed using the Multi-Class Common Spatial Pattern (CSP) method (one-versus-rest scheme), which discriminates features in feature space and improves the accuracy of classification. Then, four different features, namely channel FFT energy, mean band power, mean channel energy, and Discrete Wavelet Transform (DWT) based mean band energy features, are extracted from pre-processed EEG signals and compared to find the most suitable feature for the discrimination of four-class MI tasks. Besides, three classifiers, namely Bayesian Network classifier (Naive Bays), Linear discriminant analysis (LDA), and Linear Support Vector Machine (SVM), are compared. Performance evaluation is done on BCI competition IV dataset 2a using classification accuracy along with different performance measures calculated from a confusion matrix. The performance of the LDA classifier is found better than linear SVM and Naive Bays. The presented framework with DWT as a feature extraction technique and LDA classifier has obtained an average test classification accuracy of 80.29% over four subjects out of nine. The less computational cost of this framework makes it suitable for online Motor Imagery based BCI Systems.

**Keywords:** Electroencephalogram signals (EEG) · Motor imagery (MI) · Brain computer interface (BCI) · Discrete wavelet transform (DWT) · Common spatial pattern (CSP)

## 1 Introduction

Brain Computer Interface (BCI) set up a communication between computers and humans via the neural information of the brain activity [1]. Motor Imagery(MI) based BCI systems using EEG signals are the most promising applications [2] especially for paralyzed patients and asynchronous BCIs [3]. The prominent issue in the BCI system is the balance between accuracy and speed. Therefore

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the number of features should be minimum to keep the balance between speed and accuracy.

The spatial distribution of the MI signal is different for different imagined movements. As the frequency band of a single trial varies from subject to subject, the classification of MI tasks becomes challenging. Therefore, raw EEG signals were decomposed into spatial patterns with maximum deviation in their variance using CSP [4], which helped discriminate the MI tasks. Later on, in BCI application, information transfer rate was improved using multi-class CSP method (one vs rest scheme) by Dornhege et al. [5].

The feature extraction technique contributes in obtaining better classification accuracy. Different feature extraction techniques were proposed in the literature to enhance MI signals to obtain better classification rates. Vega-Escobar et al. [6], decomposed MI signals using DWT and extracted PSD features to classify two class MI tasks. It was observed that the proposed method gave better classification accuracy than using PSD features of raw MI signals. Marcin et al. [28], used Higher Order Statistics of Wavelet Components for EEG data classification. Xu BG et al. [29], used wavelet coefficient statistics along with $6^{th}$ order AR coefficients to form the feature sets. Sakhavi et al. [7], extracted temporal features after the application of FBCSP to train CNN to classify MI signals. Yang et al. [8], used augmented CSP (ACSP) features to classify MI signals. This reduces the loss of the important frequency information when fixed bands are selected for feature extraction. Lu et al. [9], used frequency domain features extracted from EEG signals using Fast Fourier Transform (FFT) and wavelet package decomposition (WPD) to train three restricted Boltzmann machine (RBM). It was found that FFT performed better than WPD. Hardik et al. [10], identified and removed artifact present in the data during pre-processing stage using modified common spatial pattern (CSP), which had led to better CSP log variance features and used Self-Regulated Interval Type-2 Neuro-Fuzzy Inference System (SRIT2NFIS) for classification of Motor Imagery tasks. Sakhavi et al. [11], extracted energy-based features using FBCSP and fed to convolution neural networks (CNN). It was found that the proposed method performed better than Support Vector Machine (SVM) with static energy features significantly. Thang et al. [12], selected optimal time slice and components to train CSP filters and discriminated the mental tasks by using Event Related Desynchronization/ Event Related Synchronization (ERD/ERS) effects. Hamedi et al. [13], extracted Root Mean Square (RMS) features and Integrated EEG (IEEG) and compared Multilayer Perceptron and Radial Basis Function Neural Networks for the MI tasks discrimination. Xygonakis et al. [14], extracted CSP variance features from the selected regions of interest (ROIs) using CSP filters and achieved better classification accuracy using LDA for four-class MI signals. Resalet et al. [15], revealed that the alpha Band Pass (alpha-BP) feature using power spectral density (PSD) was the best feature in MI-based applications after comparing with alpha band power fast fourier transform (alpha-BP-FFT), Mean Absolute Value and Auto-Regressive (AR) for 3-class MI data. Mohamed et al. [16], used Continuous Wavelet Transform (CWT) to extract features from MI signals, and the

classifiers, namely SVM, RBFN, and Bayes Net, were employed to discriminate left, right-hand movement, and forward imagery tasks. Out of three classifiers, Bayes Net was found better than the other two classifiers. In recent years, deep learning networks have outperformed to discriminate the MI tasks [11,17,18]. But the only drawback with them is the longer training time to train the networks.

Thus, from the literature, it is observed that there is a research scope to employ different combinations of features and classifiers for the multi-class classification of MI signals.

In this paper, four different feature extraction techniques, namely channel FFT energy, mean band power, mean channel energy, and mean band energy features extracted using Discrete Wavelet Transform (DWT), are compared to classify four-class MI signals. Materials and methods are described in Sect. 2 of this paper, while Sect. 3 defines the terms used for performance evaluation. Section 4 explains the results obtained, and at the end, conclusions are drawn from this work in Sect. 5.

## 2 Materials and Methods

### 2.1 Data Acquisition and Datasets

In this paper, dataset 2a by Brunner et al. from BCI Competition IV has been used [19]. Nine subjects were involved in the data acquisition process. It includes four MI classes (Right hand, left hand, both feet, and tongue movement imagination). For each subject, two sessions were recorded. Each session has six runs, and each run has 48 trials. Forty-eight trials include 12 trials of each class. So each session consists of 288 trials. Figure 1 (a) shows the data acquisition format of one trial with respect to time. At the beginning of each trial, a short acoustic warning tone was presented. Then a fixation cross appeared on the computer screen for 2 s. After that from t = 2 s a cue was displayed during which subject had to do one of the MI task out of four till t = 6 s. At the end of a trial, a short break lasted for 1.5–2.5 s, during which the subject was allowed to relax. Fig. 1 (b) shows the location of 22 channels. These 22 channels were sampled 250 Hz and three monopolar EOG channels and filtered between 0.5 100 Hz using a band-pass filter, and a notch filter 50 Hz was also applied. Invalid trials of EEG data were discarded before the experiments using artifact correction method described in [19] by Brunner et al.

### 2.2 Classification Framework

Block diagram of the classification framework is shown in the Fig. 2 and the overall classification process is detailed in the following sub-sections.

**Band-Pass Filtering.** First of all, the raw EEG data is filtered 7 Hz and 31 Hz using the 5-th order Butterworth finite impulse response (FIR) band-pass filter as brain activities related to MI lies within this frequency band [2].

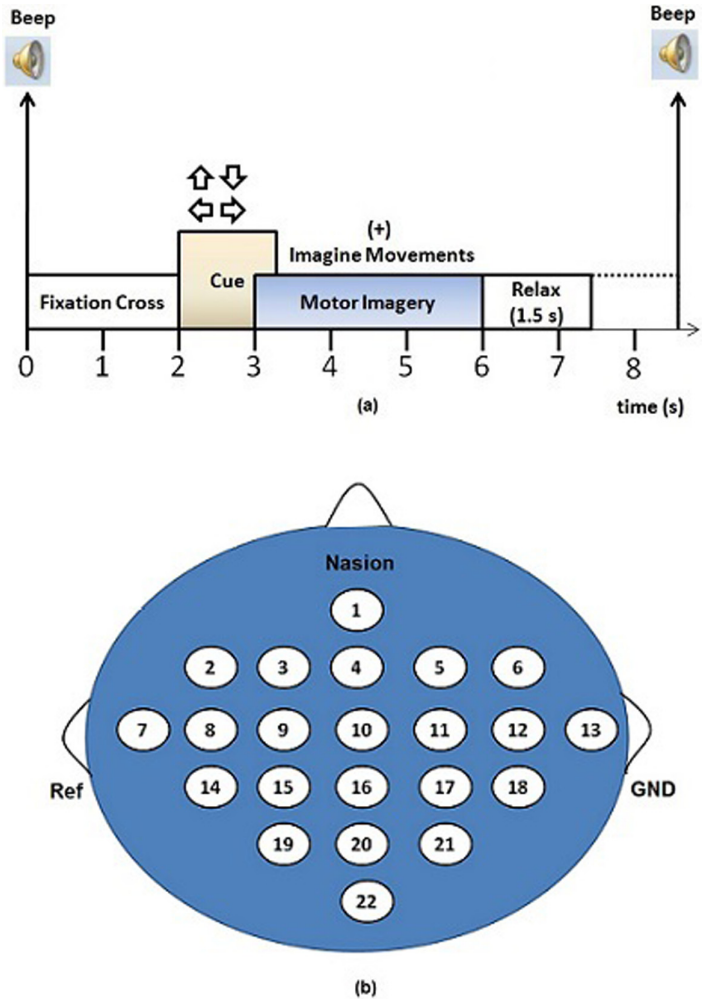



**Fig. 1.** (a) Data acquisition format of one trial with respect to time for the motor imagery paradigm; (b) Electrode placement according to the international 10–20 system (adapted from [19])

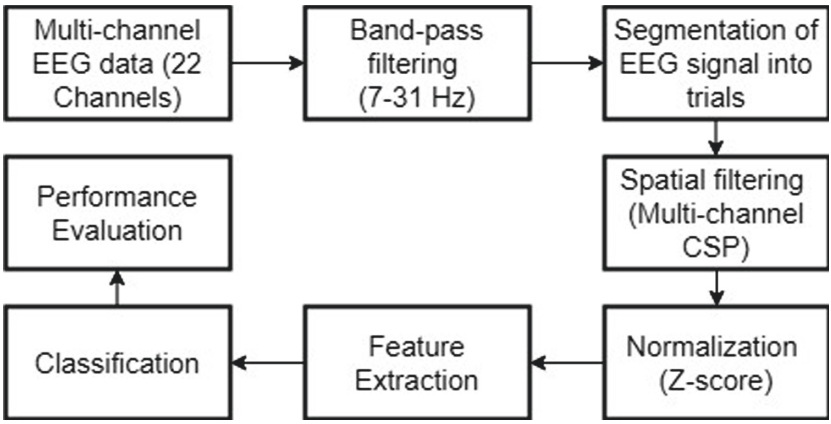



**Fig. 2.** Block diagram of the classification framework

**Segmentation.** Trials are separated from each channel using the header structure information (see [19] for details) and data from each trial is selected with the time slot starting from 2.5 s to 5.5 s.

**Common Spatial Pattern.** Common Spatial Pattern (CSP) method decomposes the raw EEG signals into spatial patterns and these spatial patterns have maximum differences in variance as shown by Naeem et al. [20]. Wang et al. in [21] concluded that this technique discriminated the classification tasks accurately.

1. Multi-class CSP:

   Consider a N*T matrix $X_i$ represents the multi-channel EEG data per trial, where i ∈ {1,2,3,4} corresponding to four MI tasks, N stands for the number of channels and T stands for the number of samples per channel. For each MI task, find average covariance matrix $\sum_i$, i∈ {1,2,3,4}. Equation (1) gives the whitening matrix as follows:

$$P = \lambda^{-1/2} {Q_0}^T \tag{1}$$

   where $Q_0$ stands for the matrix of eigenvectors and $\lambda$ stands for diagonal matrix of eigenvalues from Eq. (2):

$$\Sigma = \Sigma_{i=1}^{4} \Sigma_i = Q_0 \lambda {Q_0}^T \tag{2}$$

   To obtain the spatial filter matrix of first Class, sum of the covariance matrices of rest three classes is calculated using Eq. (3) as follows:

$$\Sigma_1^{'} = \Sigma_{i=2}^{4} \Sigma_i \tag{3}$$

following the one-versus-rest scheme [5], and if $\Sigma_1$ and $\Sigma_1^{'}$ can be translated as in Eq. (4) and Eq. (5) as

$$Y_1 = P\Sigma_1 P^T \tag{4}$$

$$Y_1^{'} = P\Sigma_1^{'} P^T \tag{5}$$

Then $Y_1$ and $Y_1^{'}$ have common eigen vectors and are given by Eq. (6) and Eq. (7) as follows:

$$Y_1 = Q_1 \lambda_1 {Q_1}^T \tag{6}$$

$$Y_1^{'} = Q_1 \lambda_1^{'} {Q_1}^T \tag{7}$$

Then the projection matrix $W_1 = Q_1^T P$ has spatial filters of the first class. similarly obtain the other four projection matrices for the remaining classes. Finally, multi-class CSP filtered signal (S) is obtained by using Eq. (8):

$$S = XW^T \tag{8}$$

where $W^T$ includes spatial filters corresponding to the four classes ($W_1$,$W_2$,$W_3$ and $W_4$) in its columns. In this work, all available 22 EEG channels are decomposed into 8 channels having maximum differences in variance as shown by [20].

**Normalization.** After pre-processing by multi-class CSP, all the trials from each channel are normalized because mean and standard deviation (SD) are sensitive to outliers. Normalization enhances features in a dataset and identifies outliers. In this step, z-score normalization [22] is used to normalize each trial as it gives the highest accuracy than other normalization techniques, such as min-max and log normalization. The z-score normalized signal S' is given by the following Eq. (9):

$$S' = \frac{S - \mu}{\sigma} \tag{9}$$

where $\mu$ is the mean of the samples of a trial, $\sigma$ represents the standard deviation of the samples.

**Feature Extraction.** Eight channels (obtained after decomposition using Multi-class CSP) are used in this step and the classification of MI tasks is carried out using the following feature extraction techniques:

1. DWT based Mean Band Energy (MBE)
   A four-level decomposition of a signal using DWT is shown in the Fig. 3. DWT is a fast, non-redundant transform than Continuous Wavelet Transform [23,24]. Mother wavelet functions, namely Daubechies 2 (db2), Daubechies 4 (db4), Daubechies 6 (db6), and Daubechies 7 (db7), are compared, and dB7 mother wavelet function is selected as it gives the best feature set. Also, DWT with four-level decomposition and Debauchies 7 (db7) [25] as a mother wavelet is used to decompose each trial from the selected channels to get the approximation and detailed coefficients.

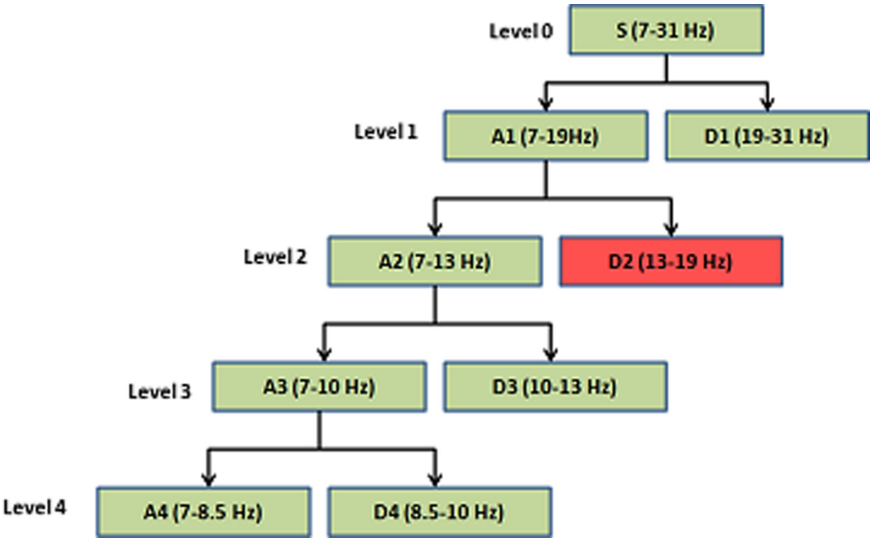



**Fig. 3.** Four level decomposition of a single trial using DWT

x(n) is a discrete time signal, where n = 0,1,2,. . . , M-1. DWT coefficients for high pass filters and low pass filters are calculated using Eq. (10) and (11) respectively.

$$W_\phi(j_0,k) = \frac{1}{\sqrt{M}} \Sigma_n x(n) \sqrt{2} \phi_{j_0,k}(n) \tag{10}$$

$$W_\psi(j,k) = \frac{1}{\sqrt{M}} \Sigma_n x(n) \psi_{j,k}(n) \tag{11}$$

where, j $\geq j_0$, $W_\phi$ and $W_\psi$ are approximation and detailed coefficients respectively, while $\phi_{j_0,k}$(n) is given by Eq. (12) as

$$\phi_{j_0,k}(n) = \sqrt{2}\phi(2^{j_0}n - k) \tag{12}$$

and $\psi_{j,k}$(n) is given in Eq. (13):

$$\psi_{j,k}(n) = \Sigma_r h_\psi(r) \sqrt{2} \phi(2^j n - r) \tag{13}$$

where $\phi$(x) is a scaling function and $\psi_{j,k}$(n) is a wavelet function

Generally, $j_0 = 0$, j = 0, 1, 2, . . . , J-1, $M = 2^J$ and k is the shift parameter. In the Fig. 2, J = 6 i.e. six level decomposition. Discrete signal x(n) is reconstructed using IDWT as in Eq. (14):

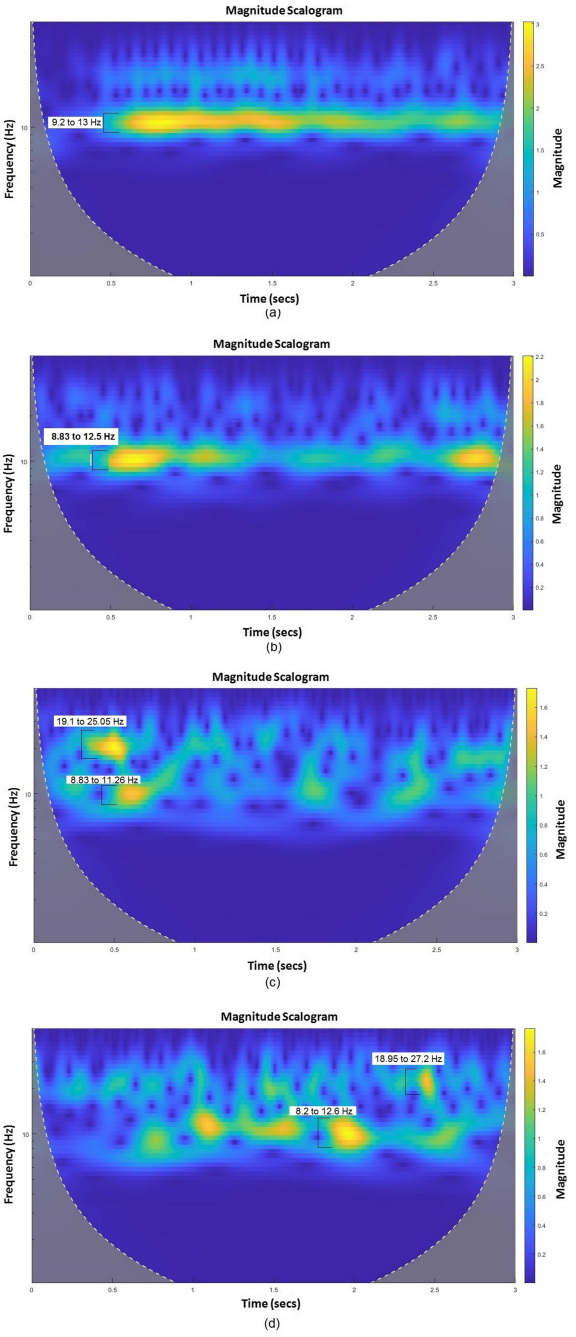



**Fig. 4.** Magnitude scalogram representation for (a) left hand, (b) right hand, (c) both feet and (d) tongue Motor Imagery

$$x(n) = \frac{1}{\sqrt{M}} \Sigma_k W_\phi(j_0, k)\phi_{j_0,k}(n) + \Sigma_{j=0}^{\infty} \Sigma_k W_\psi(j, k)\psi_{j,k}(n) \quad (14)$$

where $W_\phi$ and $W_\psi$ are approximation and detailed coefficients respectively. Also the reconstructed x(n) can be using Eq. (15) (see Fig. 3 for details):

$$x(n) = A_4(n) + D_4(n) + D_3(n) + D_1(n) \quad (15)$$

where $A_n$ and $D_n$ are approximation and detailed coefficients at level 'n' respectively.

Finally, band-energy of A4, D4, D3, and D1 is calculated (except D2 from 13–19 Hz), and their mean is taken as DWT band energy feature (see Eq. (17)). These four bands are selected as a percentage of energy for the coefficients within these bands is higher for the MI tasks (see Fig. 4). Magnitude scalogram is drawn using Continuous Wavelet Transform (CWT) with Morse wavelet.

The energy of each band is computed as the mean of squared time domain samples using Eq. (16):

$$BP_i = \frac{1}{N} \Sigma_{k=1}^{N} x_i[k]^2, i = 1, 2, \ldots, 4. \quad (16)$$

where $x_i$ corresponds to the $i^{th}$ band with k = 1,2, ..., N samples and $BP_i$ is the band energy of the $i^{th}$ band of a trial decomposed using DWT. Mean Band energy is calculated using Eq. (17):

$$MBE = \frac{1}{4} \Sigma_{i=1}^{4} BP_i, i = 1, 2, \ldots, 4. \quad (17)$$

where MBE represents Mean Band Energy and these values are then used as feature vector components.

2. Mean band power (BP)

The power of three major frequency bands: 8–14 Hz, 19–24 Hz and 24–31 Hz of each trial is calculated. The power is obtained by squaring the resulting signal and the signal is filtered using a w-sized smoothing window operation as per Eq. (18).

$$p'[n] = \ln\left(\frac{1}{w} \sum_{k=0}^{w} p[n-k]^2\right) \quad (18)$$

The mean power values are calculated using Eq. (19):

$$y_i = \frac{1}{N} \sum_{k=1}^{N} p'[k], \overline{1, n} \quad (19)$$

where $y_i$ is the mean power value of each computed band. These values are then used as feature vector components [26].

3. Channel FFT Energy (CFFT)

   As analyzed by Cecotti et al. in [26], this method employs the Fast Fourier Transform (FFT) for computing i-th EEG channel signal energy estimation in the frequency domain. The FFT result is squared and the sum of all elements is computed (20):

$$y_i = \log\left(\sum_{k=1}^{N} FFT(x_i)^2\right)\overline{1,n} \tag{20}$$

   Final feature vector components are formed after Box-Cox transformation is applied.
4. Mean Channel Energy (MCE)

   The energy of each i$^{th}$ EEG channel is computed as the mean of squared time-domain samples using Eq. (21). The result is then transformed using a Box-Cox [27] transformation (i.e., logarithm) to make the features more normally distributed. Finally, the resulting values are combined into a feature vector.

$$y_i = \log\left(\frac{1}{N}\sum_{k=1}^{N} x_i[k]^2\right)\overline{1,n} \tag{21}$$

   where N stands for the number of samples in a channel and n represents the total number of channels.

**Classification.** In this step, extracted feature sets from nine datasets of session 1 (A01T to A09T) are used to train the classifiers using 10-fold cross-validation. The session two datasets are tested using this trained classifier to obtain session-to-session transfer classification accuracy. Three classifiers, namely Bayesian Network classifier, Linear Support Vector Machine (SVM), and Linear Discriminant Analysis (LDA), are employed to classify four MI tasks.

## 3 Performance Evaluation Metrics

The performance of the proposed framework is evaluated using the following performance measures.

**Confusion Matrix:** A confusion matrix is used to evaluate the performance of a classification model.

1. **Classification Accuracy.** Classification accuracy is calculated for the performance evaluation of the classification framework with different feature extraction techniques. It is the percentage of correct decisions made in classifying the total tasks and is defined as:

$$Accuracy = \frac{(TruePositive + TrueNegative)}{(TotalInstances)} \tag{22}$$

2. **Precision**

$$Precision = \frac{TP}{(TP + FP)} \tag{23}$$

3. **Recall**

$$Recall = \frac{TP}{TP + FN} \tag{24}$$

4. **Specificity**

$$Specificity = \frac{TN}{(FP + TN)} \tag{25}$$

5. **Area under curve (AUC)**

$$AUC = 0.5 * (Recall + Specificity) \tag{26}$$

6. **F1 Score**

$$F1Score = \frac{2 * Precision * Recall}{(Precision + Recall)} \tag{27}$$

where True positive (TP) is the number of 'positive' patterns categorised as 'positive', False positive (FP) is the number of 'negative' patterns categorised as 'positive', False negative (FN) is the number of 'positive' instances categorised as 'negative', and True negative (TN) is the number of 'negative' patterns categorised as 'negative'.

**Table 1.** Session-to-session transfer classification accuracy (%) of all subjects from dataset 2a using four different features using LDA classifier

| Subject | BP | MCE | CFFT | MBE |
|---|---|---|---|---|
| S1 | 75.15 | 74.80 | 74.80 | **76.73** |
| S2 | 35.74 | **43.16** | **43.16** | 41.31 |
| S3 | 77.38 | **79.86** | **79.86** | **79.86** |
| S4 | 51.63 | 56.89 | 56.89 | **57.98** |
| S5 | 37.04 | **32.69** | **32.69** | 32.63 |
| S6 | 43.65 | 46.90 | 46.90 | **47.91** |
| S7 | 81.31 | 82.03 | **82.39** | 82.29 |
| S8 | 49.55 | 62.83 | 62.83 | **63.88** |
| S9 | 77.40 | 81.57 | 81.57 | **82.29** |
| Mean | 58.76 | 62.31 | 62.35 | **62.76** |

## 4 Results and Analysis

In this paper, classification of the four-class (Left hand, right hand, feet, and tongue) MI signals is performed using four different feature extraction techniques. The overall classification framework has been designed in Matlab R2018a

**Table 2.** Classification accuracy of each class for all subjects using MBE features. L, R, F and T represent Left, Right, Feet and Tongue classes. Ovr stands for the overall accuracy.

| Subject | LDA | | | | | Linear SVM | | | | | Naive Bays | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | L | R | F | T | Ovr | L | R | F | T | Ovr | L | R | F | T | Ovr |
| S1 | 79.16 | **84.72** | **70.83** | 72.22 | **76.73** | **86.11** | 65.27 | **70.83** | 77.77 | 75 | 66.66 | 80.55 | 54.16 | **86.11** | 71.87 |
| S2 | **31.94** | 45.83 | **44.44** | 43.05 | 41.31 | **31.94** | **58.33** | **44.44** | 40.27 | **43.75** | 19.44 | 43.05 | 34.72 | **48.61** | 36.45 |
| S3 | **87.5** | **100** | 51.38 | **80.55** | **79.86** | 81.94 | **100** | **62.5** | 66.66 | 77.77 | 73.61 | **100** | 61.11 | 66.66 | 75.34 |
| S4 | 20.83 | **81.94** | 61.11 | 68.05 | **57.98** | 16.66 | 80.55 | **62.5** | 70.83 | 57.63 | **66.66** | 43.05 | 30.55 | **80.55** | 55.2 |
| S5 | **70.83** | 2.77 | 50 | 6.94 | **32.63** | 20.83 | 4.16 | **93.05** | 12.5 | **32.63** | 2.77 | **9.72** | 76.38 | **20.83** | 27.43 |
| S6 | 66.66 | 25 | 37.5 | **62.5** | **47.91** | 62.5 | **30.55** | 34.72 | 58.33 | 46.52 | **70.83** | 12.5 | **47.22** | 40.27 | 42.7 |
| S7 | **72.22** | **98.61** | **90.27** | 68.05 | **82.29** | 62.5 | 93.05 | 83.33 | **77.77** | 79.16 | 52.77 | 84.72 | 80.55 | 72.22 | 72.56 |
| S8 | **84.72** | **29.16** | 56.94 | 84.72 | **63.88** | 83.33 | 15.27 | **61.11** | **93.05** | 63.19 | 65.27 | 8.33 | 50 | 83.33 | 51.73 |
| S9 | **90.27** | **66.66** | **79.16** | 93.05 | **82.29** | **90.27** | 56.94 | 73.61 | 97.22 | 79.51 | 88.88 | 31.94 | 48.61 | **98.61** | 67.01 |
| Mean | **67.12** | **59.41** | 60.18 | 64.34 | 62.76 | 59.56 | 56.01 | **65.12** | **66.04** | 61.68 | 56.32 | 45.98 | 53.7 | **66.35** | 55.58 |

**Table 3.** Average confusion matrix of all subjects for four classes (Values in the table are number of trials.)

| LDA | | | | | Linear SVM | | | | | Naive Bays | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Labels | L | R | F | T | Labels | L | R | F | T | Labels | L | R | F | T |
| L | **48.33** | 13.44 | 5.67 | 4.56 | L | 42.89 | 14.00 | 10.67 | 4.44 | L | 40.56 | 13.22 | 10.33 | 7.89 |
| R | 13.22 | **42.78** | 9.11 | 6.89 | R | 11.78 | 40.33 | 12.11 | 7.78 | R | 12.89 | 33.11 | 13.00 | 13.00 |
| F | 8.11 | 5.33 | 43.33 | 15.22 | F | 5.11 | 5.22 | **46.89** | 14.78 | F | 6.33 | 6.56 | 38.67 | 20.44 |
| T | 7.00 | 6.89 | 11.78 | 46.33 | T | 4.89 | 5.56 | 14.00 | **47.56** | T | 7.78 | 3.00 | 13.44 | **47.78** |

**Table 4.** Comparison based on performance measures (in %).

| Classifiers | Labels | Precision | Recall | Specificity | AUC | F1-score |
|---|---|---|---|---|---|---|
| LDA | L | 0.66 | 0.67 | 0.87 | 0.77 | 0.66 |
| | R | 0.57 | 0.59 | 0.88 | 0.74 | 0.57 |
| | F | 0.64 | 0.60 | 0.88 | 0.74 | 0.61 |
| | T | 0.63 | 0.64 | 0.88 | 0.76 | 0.61 |
| Linear SVM | L | 0.64 | 0.60 | 0.90 | 0.75 | 0.61 |
| | R | 0.67 | 0.56 | 0.89 | 0.72 | 0.54 |
| | F | 0.62 | 0.65 | 0.83 | 0.74 | 0.62 |
| | T | 0.62 | 0.66 | 0.88 | 0.77 | 0.62 |
| Naive Bays | L | 0.58 | 0.56 | 0.88 | 0.72 | 0.56 |
| | R | 0.59 | 0.46 | 0.89 | 0.68 | 0.46 |
| | F | 0.60 | 0.54 | 0.83 | 0.68 | 0.54 |
| | T | 0.51 | 0.66 | 0.81 | 0.74 | 0.58 |

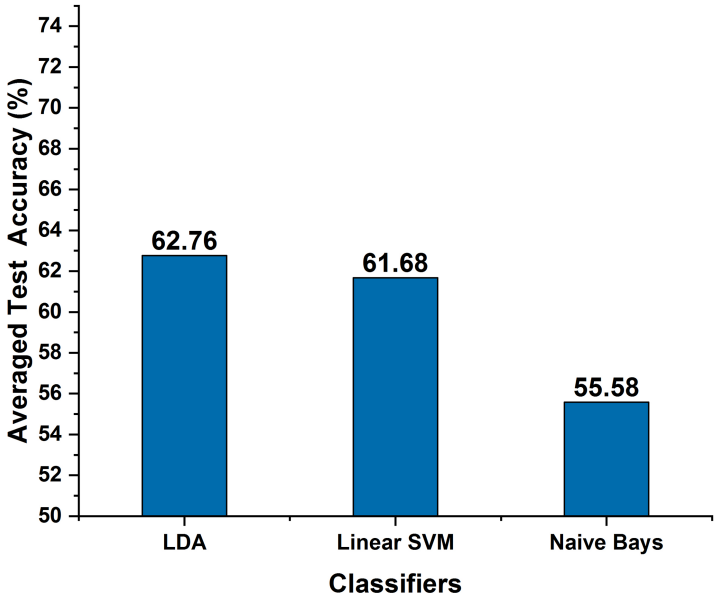



**Fig. 5.** Comparison between three classifiers showing averaged test classification accuracy (session 2)

on windows 8.1 operating system with 4 Gb RAM and evaluated on dataset 2a of BCI competition IV. Four different features, namely BP, MCE, CFFT, and MBE are calculated and compared. The performance evaluation is carried out by calculating classification accuracy. Results has been detailed in Table 1. It is observed that the framework with MBE features have been giving the best averaged test classification accuracy in comparison with the other three features (see Table 1).

In addition, Fig. 5 shows the comparison of three classifiers, namely Naive Bays, Linear SVM, and LDA, for a given framework. It is found that LDA has been giving the best averaged test classification accuracy of 62.76% for the session 2 dataset. Furthermore, the framework with DWT feature extraction method has obtained an averaged test classification accuracy of 80.29% over four subjects (S1, S3, S7, and S9) out of nine (see Table 1), which makes it suitable for online Motor Imagery based BCI Systems.

Table 2 shows the contribution of the LDA classifier in the overall classification accuracy compared with Linear SVM and Naive Bays classifier. For this, the classification accuracy of each class is calculated for all the subjects using MBE features with three classifiers (see Table 2). The ratio of the correctly classified trials for one class to the total number of trials in that class represents the Class classification accuracy. Also, the average confusion matrix over all subjects is shown in Table 3. Table 2, shows the reliability of the LDA classifier in classifying right hand and left-hand class without compromising the classification of

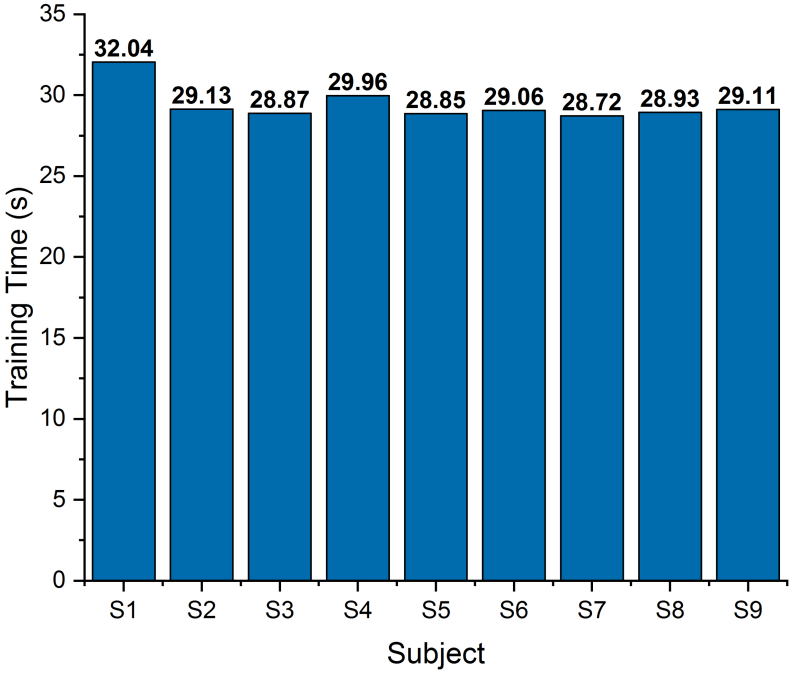



**Fig. 6.** Time required by the classification framework to train LDA classifier using MBE features for BCI competition IV dataset 2a (session 1)

feet and tongue classes. The confusion has been reduced between left/right and feet/tongue classes (see Table 3).

In order to select the best classifier for the proposed framework, the performance metrics are calculated for LDA, Linear SVM, and Naive Bays classifier (see Table 4). A good model should have higher precision, recall, specificity, Area under Curve (AUC), and F1 score, i.e., it should have low FP and low FN. Results show that LDA has performed better than Linear SVM and Naive Bays.

Furthermore, the time required by the proposed framework to train LDA classifier using DWT based MBE features for each subject (session 1) is shown in Fig. 6. An average time of 29.4 s is required to train the LDA classifier to classify four MI tasks. Such a less computation time of the presented framework makes it suitable for online Motor Imagery based BCI systems.

## 5 Conclusion

A framework is presented for the discrimination of four-class Motor Imagery signals, and its performance has been evaluated using publicly available BCI Competition IV dataset 2a. This framework with DWT based feature extraction technique has been performed better than the other feature extraction techniques

(BP, MCE, and CFFT). The presented classification framework with LDA classifier with 10-fold cross-validation has achieved an averaged test classification accuracy of 80.29% over four subjects out of nine from BCI Competition IV dataset 2a. In addition, the proposed framework's performance evaluation is carried out using different performance measures calculated using a confusion matrix. Results show that the LDA classifier has been giving better results to discriminate the four MI tasks in comparison with Naive Bays and Linear SVM. Also, the less computational cost of the framework with MBE features makes it suitable for real time applications.

It could be possible to modify the CSP method to get the best results for all nine subjects, which would improve the mean classification accuracy for all subjects; this approach will be extended in the future.

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